

Physics-constrained graph modeling for building thermal dynamics

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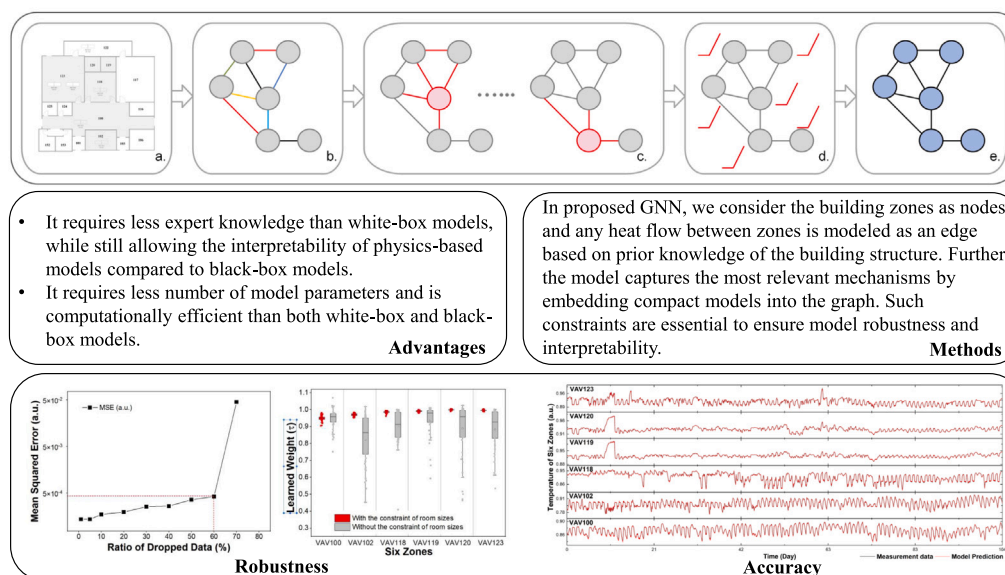
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GRAPHICAL ABSTRACT

Physics-constrained Graph Modeling for Building Thermal Dynamics



HIGHLIGHTS

- We develop a Graph-based modeling approach for building thermal dynamics.
- It requires a little expert knowledge and is computationally efficient.
- It allows the interpretability of white-box models compared to black-box models.

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ABSTRACT

In this paper, we propose a graph model embedded with compact physical equations for modeling the thermal dynamics of buildings. The principles of heat flow across various components in the building, such as walls

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Graph Neural Networks
Compact model
Building thermal dynamics

and doors, fit the message-passing strategy used by Graph Neural networks (GNNs). The proposed method is to represent the multi-zone building as a graph, in which only zones are considered as nodes, and any heat flow between zones is modeled as an edge based on prior knowledge of the building structure. Furthermore, the thermal dynamics of these components are described by compact models in the graph. GNNs are further employed to train model parameters from collected data. During model training, our proposed method enforces physical constraints (e.g., zone sizes and connections) on model parameters and propagates the penalty in the loss function of GNN. Such constraints are essential to ensure model robustness and interpretability. We evaluate the effectiveness of the proposed modeling approach on a realistic dataset with multiple zones. The results demonstrate a satisfactory accuracy in the prediction of multi-zone temperature. Moreover, we illustrate that the new model can reliably learn hidden physical parameters with incomplete data.

1. Introduction

Nowadays, energy consumption in buildings accounts for around 40% of worldwide energy consumption, with the majority of energy used for heating, ventilation, and air conditioning (HVAC) [1,2]. Thus energy-efficient buildings have become one of the top priorities for addressing global energy sustainability and decarbonization objectives [3]. Advanced building control strategies have been demonstrated to save up to 28% of energy [4–6], in which thermal load prediction plays an essential role in HVAC control [7,7–13].

White-box models are proposed firstly, which are grounded in first principles and thus equipped with the characteristic of the interpretability. Such physics-based models rely on solving Ordinary Differential Equations (ODEs), such as convection, and conduction equations, describing building thermal dynamics to calculate the zone temperature [14]. When properly calibrated, White-box models can generate robust and accurate predictions [15]. However, real-world building dynamics are high-dimensional, nonlinear, and often uncertain, for which white-box modeling typically requires special expertise and long development time [15,16]. Moreover, despite the significant effort in development, white-box models can still make inaccurate predictions because of multiple sources of uncertainty [17]. Furthermore, solving such white-box models causes high computation costs, thus inefficient in various applications [18,19].

As more computationally efficient alternatives, data-driven approaches with black-box models have been widely used in the analysis of dynamic systems [20–25]. Black-box models rely on patterns from historical data to achieve high accuracy without expert knowledge required. They are flexible and scaled to large systems straightforwardly, like Multiple Linear Regression (MLR), Support Vector Regression (SVR), and Neural Networks (NNs). [26] applies recurrent NNs (RNNs) to predict energy consumption. In [27], the paper proposes a specific gated Convolutional NN (CNN) with the result that it outperforms RNNs on day-ahead multistep hourly predictions of the electricity consumption. However, purely data-driven black-box models cannot make predictions outside the training data regime, where unrealistic control signals can impair overall control performance [28,29]. Furthermore, the data-driven models require a large volume of data, making data collection too expensive in some cases, and invisible but essential data not accessible in some other cases, easily casting complex models as overfitting on small datasets [17,28,29]. More importantly, the black-box models are challenging to be interpreted and generalized, implying that they contribute little to understanding the dynamic system's behavior [30].

Considering a trade-off between model accuracy, computational efficiency, and interpretability, the gray-box models capture the most relevant mechanisms by simplifying the building thermal dynamics and introducing model parameters to be learned from data. For example, [31–33] simplify the model to reduced order resistance (R) and capacitance (C) models. The simplified model requires less expert knowledge than white-box models while still allowing the interpretability of physics-based models. Because they are effective, we rely on the reduced RC models as the basis to construct our model. However, most

reduced data-driven approaches are oversimplified, applied to single-zone buildings, and typically minimize only 1-step ahead prediction errors [34,35]. For a large-scale building with multiple zones, one may estimate the parameter values in resistance and capacitance models from the measured data, which requires heuristic tuning and increases the development time and cost [36].

Additionally, the integration of classical physics-based models with the latest machine learning techniques is increasingly popular in recent years to model the dynamic systems [37–39]. That is because the physical-constrained model minimizes expert knowledge and training data while retaining the interpretability characteristic of physics-based models, thereby aiding in understanding the behavior of dynamic systems. Moreover, the physical constraint model can fit data well even when uncertainties are not well-described in physics. The general physics-informed neural network (PINN) [40] approximates hidden physical parameters by a deep neural network and enforces physical constraints by augmenting the loss function via regularization and penalty. However, learning many hidden parameters requires an extensive neural network, thus increasing the computation cost. Most physical-constrained neural network impose physical constraints by loss function augmentation via regularization and penalty methods [41–43]. However, the increase in the number of penalty terms in the loss function further causes the trade-off between multiple tasks [44,45], which is difficult to tune. In a recent excellent work, [37] applied PINNs to building models. They leveraged known physical rules to enforce constraints via penalty and used the Perron–Frobenius theorem to limit the stability and dissipativity of the system by bounding the eigenvalues of all the NNs. [38] processed part of the inputs separately by a linear module, inspired by physics-based RC models. Such a structure ensures thermal physics, such as the positive correlation between input power and zone temperatures. In this paper, we embed the physics-based model into the graph to enforce the physics.

In this work, we introduce a novel data-driven modeling approach for building thermal dynamics, meticulously balancing model accuracy, computational efficiency, interpretability, and robustness. The proposed model minimizes expert knowledge and training data while retaining the interpretability characteristic of physics-based models. This is achieved by embedding compact models into the nodes and edges of the graph, effectively capturing the behavior of each component. Additionally, we demonstrate the model's accuracy and robustness to incomplete data.

The organization of the rest of the paper is as follows: In Section 2, the nodal regression task is formulated, and the proposed model is described in detail. The effectiveness of the proposed model is validated in Section 3 with extensive comparisons and visualizations. Finally, Section 4 concludes the paper.

2. Model development

2.1. Problem definition and overview

In this section, we provide a brief description of the building thermal data and formalize the state estimation problem as a M-steps ahead GNN prediction task.

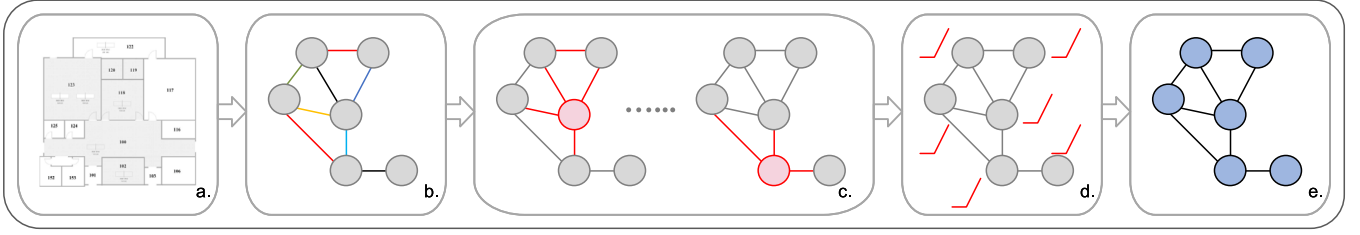


Fig. 1. Overview of the graph modeling flow. a. The original structure of the dynamic systems. b. The graph representation of the dynamic system, including the components (nodes) and their interactions (edges). Neighboring components may have different correlations, shown as different colors in the figure. These correlations can be modeled as differential equations, compact models or neural networks. c. The GNN approach integrates physical models and neural networks to model the system. Two main steps, aggregation and transformation, are applied to each node, which aggregate the features from neighboring nodes and transform them to a new feature space, respectively. d. The update function applied to the nodes to update their states. e. A new graph is generated from one layer of GNN, with the same graph structure but new states. Depending on the task, the cycle of aggregation/transformation and update (i.e., one layer in GNN) may be applied several times until the final result. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

Abstract parameters in the proposed model.

Neural model	Building physics	Units
Inputs	Mass flows, supply temperatures, ambient temperatures, zoom temperatures	[kg/s, K, K]
Hidden states	The average temperature change rate, heat flows	[K/s, W]
Outputs	Zoom temperatures	[K]

The data used in this study is captured in a commercial structure in Richland, Washington, that was previously described in [46]. Four air handling units (AHUs) in this building serve 24 zones, constituting a variable air volume (VAV) system. Over 600 sensors are installed throughout the building to collect data and send them to the database, which is connected to the building management system (BMS) with a one-minute resolution. In practice, sensor data from the natural environment is usually noisy and incomplete, posing a significant challenge to the robustness of the algorithm. The raw data is pre-processed to mitigate this issue, as described in [46].

For our modeling purpose, the thermal dynamics of a multi-zone building are represented by an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, v_2, \dots, v_n\}$ denotes the indexed vertex set with $|\mathcal{V}| = N$, and $\mathcal{E}$ is the edge set with $|\mathcal{E}| = E$. Vertices in the graph correspond to zones of the building. When there exists a heat transfer between two zones, there is an edge in $\mathcal{E}$ connecting the corresponding vertices.

We assign a set of dynamics features, $d_v \in N$, to each vertex that varies with time. Table 1 summarizes the parameters used for the proposed model. Furthermore, we assign trainable static features to each vertex representing time-invariant characteristics. For instance, we assume that the average temperature change rate, defined as τ , depends on its intrinsic properties only, such as size and location. The feature matrix $\mathcal{X}^{(t)} \in \mathcal{R}^{N \times (d_v+1)}$ concatenates all these features together at time t . The dynamic states of $\mathcal{G}$ at time t is represented by the zone temperature $\mathcal{Z}_v^{(t)}$ at each vertex $v \in \mathcal{V}$.

Given the graph data representation shown in Section 2.2, the state estimation (SE) corresponds to a supervised node regression task. Specifically, SE aims to infer zone temperature $\mathcal{Z}_v^{(t)}$ at time t from previous l states ($\mathcal{Z}_v^{(t-1)}, \mathcal{Z}_v^{(t-2)}, \dots, \mathcal{Z}_v^{(t-l)}$) and measurements $\mathcal{X}^{(t)}$. Mathematically, we define a general one-layer GNN and a node regression output as:

$$w^{(t)}(e) = |f_e(\mathcal{Z}_u^{(t-1)}, \mathcal{Z}_v^{(t-1)})| \quad (1)$$

$$H_v^{(t)} = f_h(\text{Agg}(v), \mathcal{X}_v) \quad (2)$$

$$\text{Agg}(v) = \sum_{e \text{ out of } v} (\mathbb{I}(r) - \mathbb{I}(s))w(e) \quad (3)$$

$$\mathcal{Z}_v^{(t)} = f_v(H_v^{(t)}, \mathcal{Z}_v^{(t-1)}, \mathcal{Z}_v^{(t-2)}, \dots, \mathcal{Z}_v^{(t-l)}) \quad (4)$$

where edge $e = (u, v) \in \mathcal{E}$, and nodes $u, v \in \mathcal{V}$. $w^{(t)}(e)$ refers to the value of the (u, v) entry, corresponding to edge e of the adjacency matrix $\mathcal{W}$. f_e is a non-linear function transforming the nodal state vectors at

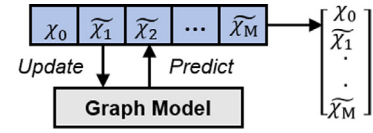


Fig. 2. Direct time stepper to update the initial condition based on the current states and predict M time steps through the graph model [47]. This update-prediction process continues with all data.

the ends of edge e to their relationship. f_h transforms concatenated node features of its attributes and aggregated features from its neighbor nodes to the embedding $H_v^{(t)}$. $\text{Agg}(v)$ represents an aggregation function that pools information of each node's neighborhood defined in $\mathcal{E}$. Since heat is always transferred from the object at the higher temperature to the lower temperature, we assume that the object at the higher temperature is the sender and the object at the lower temperature is the receiver. $\mathbb{I}(r)$ is the indicator function: $\mathbb{I}(r) = 1$ if the node is the receiver, else $\mathbb{I}(r) = 0$. Similarly, $\mathbb{I}(s)$ is the indicator function to determine whether the node v is the sender. f_v as the update function aims to update the temperature of the node based on the embedding at time t . We follow the model definition of the direct time stepper to predict a sequence of states, as discussed in [47]. The model is designed to predict future states for the next M time steps based on previous states. In this study, each time step represents a 15-min interval, consistent with the regular 15-min spacing of the time-series data. For the prediction of these future states, we first update the initial states based on the current states, and then make the prediction. We repeat this update-prediction process for each M steps until we make all predictions, as shown in Fig. 2.

As shown in Fig. 1, given the graph data representation to be discussed in Section 2.2, each edge $e = (u, v) \in \mathcal{E}$ integrates the endpoint information from node $u, v \in \mathcal{V}$ and calculates the corresponding weights $w^{(t)}(e)$, or the interaction between u and v , at time t through f_e . If a pair of nodes have more than one interaction, we adopt different learning rules f_e for different relationships. At each iteration of the GNN, the aggregate function takes an input from the set of embedding of the nodes in v 's neighborhood and transforms the aggregated information with its features to the embedding $H_v^{(t)}$. Then the update function f_e is applied to update the states.

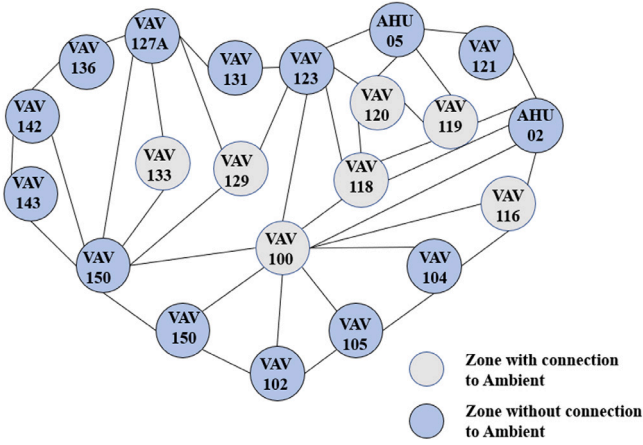


Fig. 3. Graph model of a multi-zone building. Each zone is modeled as the node and pair-wise heat transfer between zones are modeled as edges.

2.2. Graph modeling of a multi-zone building

Complex dynamic systems are generally composed of multiple components and their associated interactions. In our case, the thermal dynamics of the building are composed of multiple zones, with heat transfer occurring via conduction and convection into the zone. This process of heat flow is identical to message-passing in GNNs, which is the learning of locally-bound spectral filters. Therefore, we model zones as nodes and pair-wise heat transfer as edges, as illustrated in Fig. 3. On the physical aspect, we equip the edge with explicit physical meanings, such as heat flow rate, to improve the model's interpretability. On the aspect of model efficiency, the GNN imposes spectral constraints to the input by defining a graph, unlike the neural network that has to learn its weights from the data to account for the spatial connections from the data. In addition, heat transfer only occurs at the edge of the graph model and, thus, further reduces the solution space.

2.3. Compact thermal modeling of the nodes

In this section, we translate our update function f_e to reduced-order RC models, which capture the most relevant mechanisms while maintaining a balance of accuracy and complexity. We assume that all zones follow similar thermal mechanisms but have distinct physical parameters (i.e., trainable features with explicit physical meanings).

In practice, R and C values for building zones have to be inferred from feasible observations. However, it is difficult to derive these values from the first principles since numerous variables, such as size, location, and occupancy status, influence them. Furthermore, some invisible factors, such as material deterioration, are hard to be explicitly monitored. Therefore, extracting R and C values requires heuristic tuning, resulting in time-consuming and inaccurate estimations. In this work, we propose to define a static trainable feature, $\tau = R \times C$. Specifically, τ serves as a trainable weight, representing the average temperature change rate, rather than a constant rate at a single moment. This allows the model to produce results under varied boundary conditions and we improve results using neural networks. Its value is fitted from observable data via back-propagation.

To compensate the limitation of RC models that only consider external heat gains, we incorporate internal heat gains into our model. Fig. 4 illustrates the gained heat of zones from internal airflow provided by the variable air volume system, ambient airflow from outdoor weather, and adjacent airflow from heat exchange between zones and their neighbors. These three heat gains are combined to form a source of heating energy that charges the thermal capacitance of a zone. Through the neural network (e.g., a multi-layer perceptron), we transform three

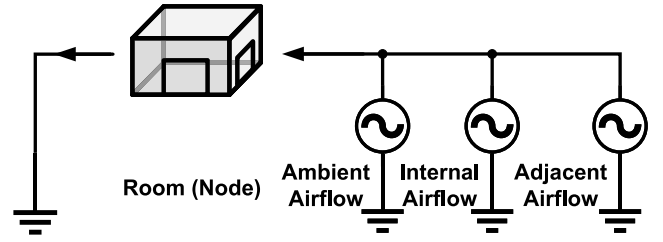


Fig. 4. Compact model of the nodes. Three major energy sources, internal airflow, provided by the variable air volume system, ambient airflow from outdoor weather, and adjacent airflow from heat exchange between zones, charge the thermal capacitance of each zone.

heat gains into an equivalent heating energy source T_s . Based on the initial condition, the compact model is then efficiently solved to generate zone temperatures in the subsequent time step:

$$Z_v^{(t)} = T_{vs}^{(t)} + (T_{v0}^{(t)} - T_s^{(t)}) \times e^{-t/\tau} \quad (5)$$

$$T_{vs}^{(t)} = f_s(H_v^{(t)}) \quad (6)$$

$$T_{v0}^{(t)} = f_0((Z_v^{(t-1)}, Z_v^{(t-2)}, \dots, Z_v^{(t-l)})) \quad (7)$$

where $Z_v^{(t)}$ is the zone temperature of node v at time step t . τ represents the rate of temperature change. f_s transforms the embedding of node v to heat energy source, and f_0 generates residual heat source $T_{v0}^{(t)}$ of node v at time step t .

Ideally, physical parameters, such as τ , should be uniquely determined for each zone. However, this is challenging in reality due to limited observable data. The proposed model will update physical parameters through iterations to mitigate this issue until the accuracy is sufficiently high. Under spectral constraints, we use the compact model to replace neural networks, which improves model physicality and reduces model parameters. Therefore, the process of model fitting is less ambiguous, helping stabilize parameter extraction, as shown in the results.

2.4. Edge modeling of the interaction

This section explains how to model the heat flow through the edge. As discussed in Section 2.1, the proposed method calculates the corresponding weights of edges based on aggregated information from their endpoints before nodal aggregation. This process is similar to that in an interaction network [48], alternatively calculating the edges' interaction and reasoning the node's behavior.

We assume that conduction and convection heat transfer are inversely proportional to thermal resistance, as the RC model. Since heat transfer shares standard thermal physics, all edges employ one interaction model for heat exchange. Mathematically, we define the interaction model across edge e as:

$$w_e^{(t)} = |(Z_u^{(t-1)}, Z_v^{(t-1)}) \times \frac{1}{R_{uv}}| \quad (8)$$

Here, $w_e^{(t)}$ represents heat flow between node u and v at time step t . T_i^{t-1} is the temperature of zone i at time step $t-1$. R_{uv} is the intrinsic thermal resistance of the associated path for heat exchange. We assume that heat exchange among zones is a relatively slow process and, thus, has a delay behind the temperature change of zones.

2.5. Loss function under physical penalty

We define the following loss function (L) with penalty terms to train the model over M time steps:

$$L = \frac{1}{M} \sum_{t=1}^M \|\bar{Z}^{(t)} - Z^{(t)}\|_2^2 + \lambda \quad (9)$$

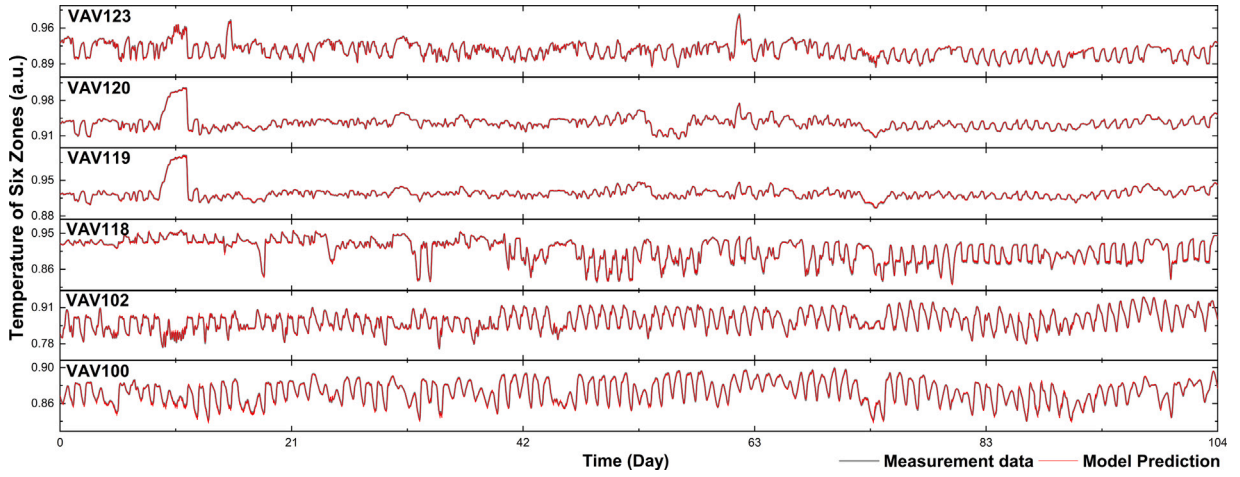


Fig. 5. The evaluation of 32-ahead predictions in six zones (VAVs). The collected data span over four months, the first three of which are training data-set, and the last month are test data.

$$\lambda = \sum_{n=1}^N \max(0, -\tau_n + \tau_{n+1}) \quad (10)$$

The first term of the loss function is the mean squared error over M time steps. The second term is used to enforce physical constraints λ , which sorts all τ s in the loss function. Because τ of each zone is approximately proportional to the area, we assume the ranking of τ s represents the ranking of zone areas, which is introduced to the loss function as a penalty term. This penalty term effectively reduces the variations in model learning, as shown in Fig. 12.

3. Results and analysis

3.1. Experimental setup

To prepare data for model training and testing, we first create a time series dataset with zone temperature, mass flow, and ambient temperature as tuples. The dataset contains 10 083 data points with a 15-min time resolution spanning four months. We replace outliers with the average of the previous ten periods' measurements. Next, we normalize all variables using the min-max method and smooth the data, as suggested in [46]. We use the first three months for training and validation and the last month for testing.

We implement the proposed graph model in the Pytorch framework [49]. With randomly initialized weights, we use the Adam optimizer with a learning rate of 0.001 to train the model. We only use one neural network to fit training data, which is a multi-layer perceptron (MLP) [50] with two layers and 16 nodes per layer from brute-force hyper-parameter search. It is the 32-ahead prediction as shown in Fig. 5. We compare the ground truth and model predictions on the test data. To evaluate the performance long-term prediction, we use the pre-trained 32-step model to predict temperatures at different steps. Fig. 6 shows that the value of mse becomes worse due to the extrapolation. That is because the long prediction causes an accumulation of errors over multiple steps without updating initial conditions.

In reality, various uncertainties may happen, leading to drastic changes of initial conditions and unpredictable behaviors that deviate from previous models. Therefore, it is necessary to update initial conditions from time to time to mitigate the uncertainties and improve the accuracy. To demonstrate this consideration, we apply the same model to simulate the thermal behavior, with or without the update of initial conditions, as shown in Fig. 7. In this experiment, we update the initial conditions per 32 steps before 144 steps, and we do not update the initial conditions after 144 steps. The result without the update presents the accumulation of errors when time increases.

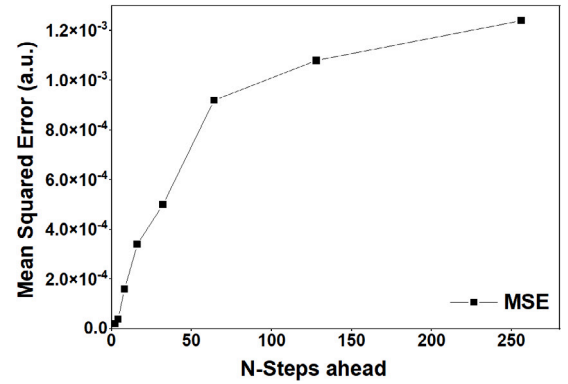


Fig. 6. We evaluate the performance of long prediction by comparing the mean squared error at each time step of the prediction horizon, where the statistics were computed from the test set.

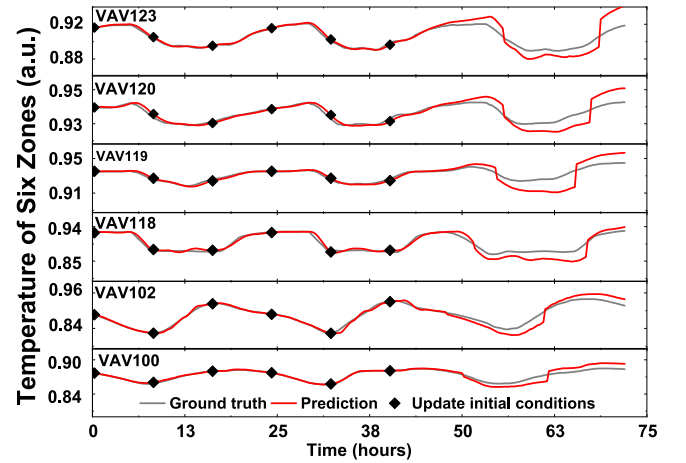


Fig. 7. Without the regular update of initial conditions, the error in model prediction accumulates.

For visual assessment of the long prediction, we set M as 288, corresponding to three days. Fig. 8 shows normalized predictions (red) of all six zone temperature trajectories with the measurements (gray).

Fig. 9 illustrates that the proposed model outperforms the reduced RC model [51] and other data-driven approaches on the same building thermal data. In this experiment, we adopt the codes of other

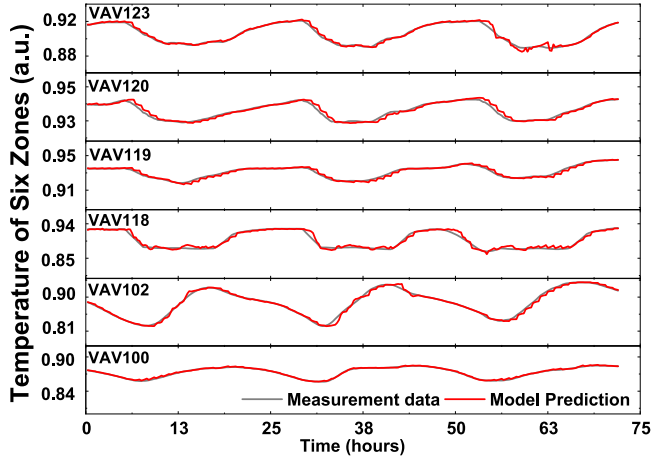


Fig. 8. Three days ahead simulation of the prediction best-performing model. Different colors represent learned (red) and measurements (gray) trajectories of the normalized temperature. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

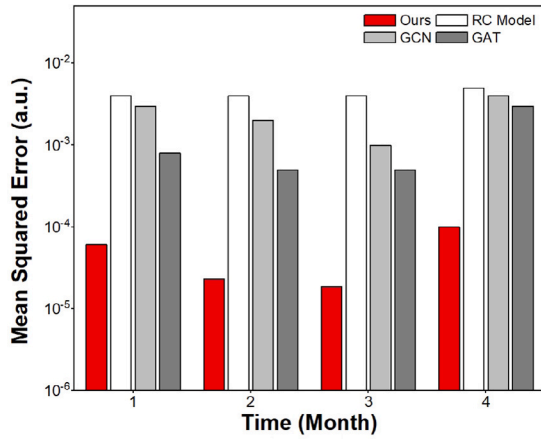


Fig. 9. Error comparison in temperature prediction across four months, between our proposed method and RC model as well as other graph-based, data-driven methods [52,53].

Graph Neural Networks (GNNs), such as Graph Convolutional Networks (GCN) [52] and Graph Attention Networks (GAT) [53], and apply them to our building data. Compared to these GNN methods that are purely data-driven, our approach integrates compact thermal models for the nodes and edges to capture the essential non-linear physics and, thus, effectively improves the accuracy. However, GNNs require a large amount of data to achieve good accuracy. The results confirm that our model outperforms others, even in small data, and all three data-driven approaches perform better with the increment of training data.

3.2. Effect of compact modeling of nodes

Mathematically the RC model has an exponential solution as its update function f_e . To demonstrate the importance of the non-linear update function, we apply the Fourier transform to expand the exponential solution and obtain the approximated first-order and third-order update functions. Compared with the exponential update function, these two approximated update functions have a much lower order of nonlinearity. In addition, we implement a alternative update functions, a pure MLP, to replace the RC model for comparison.

As illustrated in Fig. 10, the result indicates that our RC update function outperforms MLP, and other lower-order update functions, because of its better fidelity to the underlying physics. The neural network

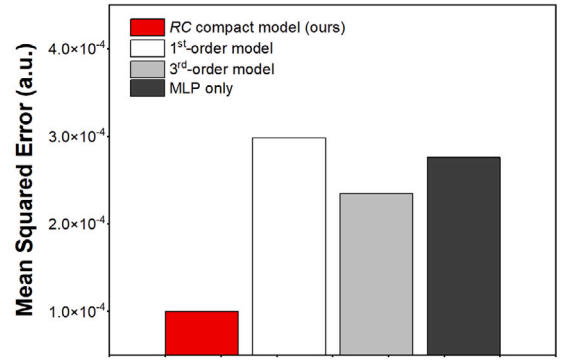


Fig. 10. The selection of update functions. To demonstrate the importance of the update function, we compare the prediction error of our proposed RC model with those from four other solutions, including two lower-order models and a black-box function with the neural network (MLP with 2 layers and 8 nodes every layer) only.

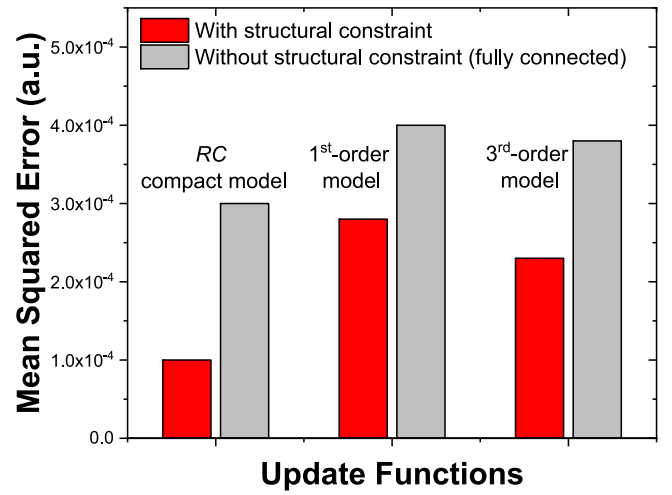


Fig. 11. The adoption of the structural constraint, i.e., the connection information between zones, improves the prediction accuracy. In this study, we compare model performance with and without structural information from the layout under RC compact model, 1st-order model and 3rd-order model, as presented in Section 3.2. Without the knowledge of the layout, we model the building as a fully connected graph to start the training.

does not fit well due to limited data. The third-order update function captures more non-linearity and, thus, better than the first-order update function.

3.3. Structural constraint on the edges

Given the layout of the building zones, we model the thermal dynamics of a multi-zone building as a partially connected graph, where nodes represent zones and edges represent pair-wise heat transfer between zones. However, without the exact layout, we would have no knowledge where heat transfer will take place. In this case, we have to assume that each zone is possibly connected to all others, which leads to a fully connected graph. In terms of model implementation, we add a trainable weight to each edge to indicate the presence of heat exchange or not. If the trainable weights exceed the threshold during model training, we then decide that heat exchange exists along that edge connecting two zones.

In this study, we compare model performance with and without the structural constraint from the layout, i.e., with a partially connected graph based on the layout and with a fully connected graph to start the model training, respectively. As expected, Fig. 11 demonstrates that the structural constraint improves performance under different update

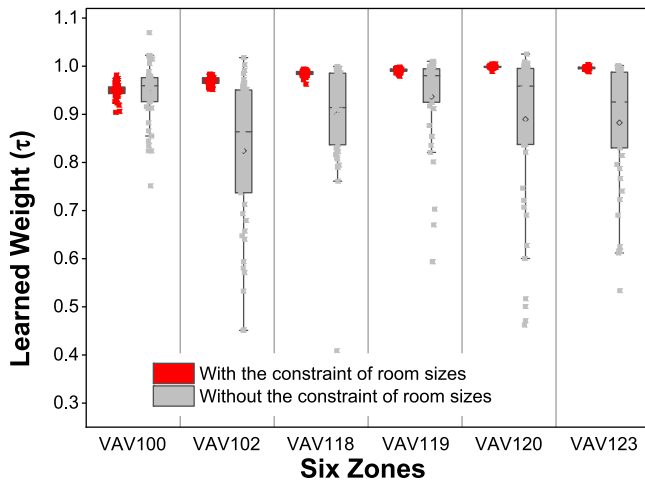


Fig. 12. Training with the physical constraint effectively reduces the distribution range of learned weights. In this study, we apply the sorting of zone size for six zones as the physical constraint to train model parameters.

functions. That is because edge weights learned from the data are influenced by other factors and uncertainties. The more constraints we can impose up front, the better we can train the model from the data.

3.4. Robustness under physical constraint

Training of GNNs is inevitably stochastic. Different model initialization leads to the convergence to different local minimum, which results in a different model solution. To study the robustness of our solution, we train the same GNN model multiple times with different initial weights. As described in Section 2.5, the sort of τ s can be added as the soft penalty term into the loss function to constrain the training. Fig. 12 demonstrates that when such a physical constraint is applied, the range of learned weights (e.g., the rate of temperature change, τ , in this example) becomes much smaller than those without the constraint, producing a more robust solution with comparable prediction accuracy.

With the physical constraint, our RC model successfully fits τ s via back-propagation and produces interpretable results. Specifically, the sort of τ s learned from collected data is VAV100 - VAV102 - VAV118 - VAV119 - VAV123 - VAV120, well matching the estimated sort based on the zone area. The temperature in VAV100 changes most slowly because the area is the biggest and open to the ambient.

3.5. Robustness under incomplete data

In reality, one of the challenges for model training is the collection of clean and complete data. Many factors may reduce the quality of the dataset, such as sensor issues, power glitch, etc. To test the robustness of our GNN model, we randomly drop raw data points from 1% to the maximum percentage of dropped data that the proposed model can tolerate. Since our model training requires consistent time series data, we linearly extrapolate the raw data before training, when there is a dropped data point. As illustrated in Fig. 13, our model can tolerate up to 60% of data missing. The reason for this is that in physical transitions are relatively smooth and consistent. Therefore, appropriate non-linear-in-time models, such as the RC model, and structural and physical constraints help keep model continuity in learning and reduce the sensitivity to data glitches.

4. Conclusion

This article presents a new learning model with compact models for non-linear thermal dynamics and the graph model for the multi-zone building structure. In a nutshell, our model combines graph

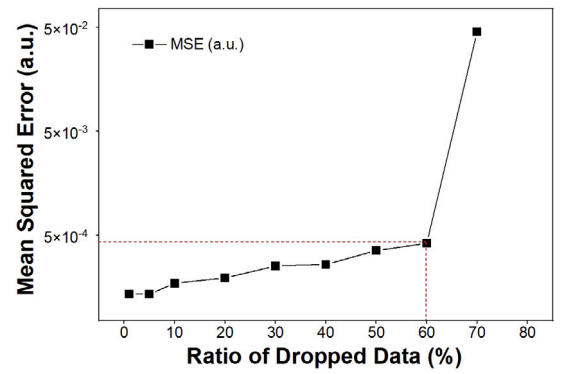


Fig. 13. Our proposed graph model with compact models is able to tolerate a large ratio of data loss.

with physical compact modeling, which is preferred in the modeling of multiple-component dynamics to describe the behavior from first principles. As a result of prior knowledge (e.g., the RC model and the connection between zones), our approach requires fewer empirical estimations, lower computational costs, and better model predictability than traditional machine learning approaches. We present accurate 288-step ahead predictions, with our best-performing model achieving a score of 0.0012 on normalized open-loop MSE. Furthermore, we interpret our model by defining and constraining physical weights and demonstrate that physical parameter τ well approximate the actual setting and the ranges of the parameter τ are markedly reduced, being up to 10 times smaller when compared to scenarios without constraints. Through experiments with realistic building data, our proposed GNN models present robust learning to tolerate up to 60% data missing. The integration of compact physical models with graph neural networks can be generalized to other dynamic systems beyond building thermal dynamics. In future work, we aim to adapt our model for more sophisticated systems and broaden its application to energy-related domains, such as heat dissipation and microelectronics.

CRediT authorship contribution statement

Ziyao Yang: Conceptualization, Formal analysis, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Amol D. Gaidhane:** Conceptualization, Supervision, Validation, Writing – original draft, Writing – review & editing. **Ján Drgoňa:** Conceptualization, Formal analysis, Methodology, Supervision, Writing – review & editing. **Vikas Chandan:** Conceptualization, Data curation, Writing – review & editing. **Mahantesh M. Halappanavar:** Data curation, Writing – review & editing. **Frank Liu:** Conceptualization, Supervision, Writing – review & editing. **Yu Cao:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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